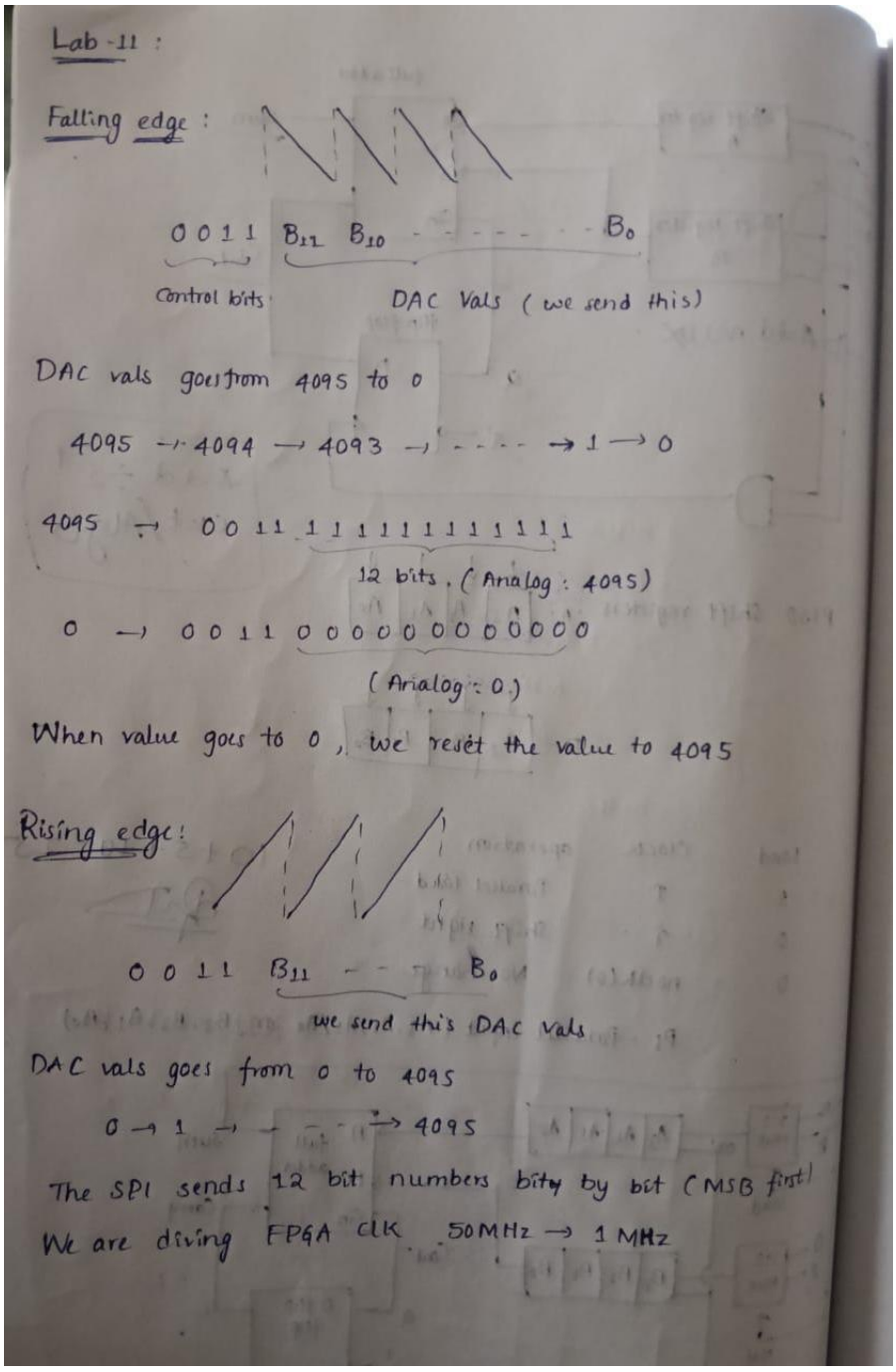
DS LAB 11 REPORT

P MAHITHA

CS24BT054

18/04/2026



Lab-11 :

Falling edge :

0 0 1 1 B_{11} B_{10} - - - - - - B_0

Control bits DAC Vals (we send this)

DAC vals goes from 4095 to 0

4095 → 4094 → 4093 → - - - - → 1 → 0

4095 → 0 0 1 1 1 1 1 1 1 1 1 1 1 1

12 bits. (Analog : 4095)

0 → 0 0 1 1 0 0 0 0 0 0 0 0 0 0

(Analog : 0.)

When value goes to 0, we reset the value to 4095

Rising edge:

0 0 1 1 B_{11} - - - B_0

we send this DAC vals

DAC vals goes from 0 to 4095

0 → 1 → - - - → 4095

The SPI sends 12 bit numbers bitw by bit (MSB first)

We are diving FPGA clk 50MHz → 1 MHz

Triangular Edge :

Logic → start from 0 & go till 4095, and on reaching 4095 → decrease the DAC value till 0

0 → 1 → - - - → 4095 → - - - → 1 → 0



10+5+10 = 25